A glycemic digital twin for the next two hours

- Happiest Health Digital Twin Challenge 2026
- Type 2 diabetes, Indian clinic units (mg/dL)
- Synthetic patients only. Not a medical device.
- Team, college, leader, phone, email: FILL IN

Why this question, in India

- ICMR–INDIAB: about 101 million people with diabetes, about 136 million with prediabetes (Anjana et al., Lancet Diabetes & Endocrinology, 2023)
- Rice- and wheat-based meals make the post-meal rise a daily event
- Follow-up is often an HbA1c every few months; CGM is still mostly out of pocket
- A district clinic or tele-endocrinology review needs a 2-hour answer, not another 3-month average

The decision the twin is allowed to make

- Primary label: CGM maximum in the next 2 hours ≥ 180 mg/dL
- Secondary label: CGM minimum in the next 2 hours < 70 mg/dL
- Inputs: the static chart, the last 6 hours of sensors, the meal and dose log
- Outputs: calibrated probability, a 4-point trajectory with a residual band, drivers, one counterfactual

What this prototype is not

- Not trained on any real electronic health record or CGM export
- Not a dosing calculator and not a closed-loop controller
- Genetic fields are simulated allele counts, not sequenced genotypes
- A tight error on synthetic glucose is not clinical accuracy

Synthetic chart: an Indian outpatient file

- 320 people for the reported run, seed 2026. Demo panel uses a different seed.
- Age, sex, city and state, BMI from a distribution centred near 27
- HbA1c mixture: most people near a treated target, a smaller group well above it
- Medicines: metformin, sulfonylurea, insulin total daily dose. Labs in mg/dL.
- Simulated TCF7L2 and CDKAL1 allele counts and a family-history flag

The wearable stream is coupled to that chart

- 5-minute CGM, heart rate, RMSSD, steps, sleep stage
- One glucose state: meals, dawn, short sleep, walking, metformin, sulfonylurea, insulin
- A logged meal raises glucose later if sensitivity is low; insulin without that meal can fall below 70
- CGM is the state plus autocorrelated noise. Labels use the CGM.

Train, validation, test: by patient

- Seed 2026: 320 patients × 3 days
- Windows — train 28672, validation 6144, test 6144
- Patients — 224 / 48 / 48
- Test prevalence of a 2-hour rise to ≥180: 0.3185
- Decision every 30 minutes. Isotonic calibrators see validation rows only.

Four models, same label

- Persistence: logistic regression on the latest CGM value
- Static: L2 logistic regression on the chart, no clock
- Wearable: histogram gradient boosting on 6-hour sensor aggregates
- Fusion: the same booster on chart + sensors + meal and dose log, then isotonic calibration
- Logged walks and skipped meals are upweighted 3× in the fusion fit. Test metrics are not.

Held-out comparison

- Persistence — AUROC 0.8355, AUPRC 0.7910, sensitivity 0.7251 at specificity 0.8073
- Static logistic — AUROC 0.7618, AUPRC 0.5968, sensitivity 0.5764 at specificity 0.8108
- Wearable — AUROC 0.9477, AUPRC 0.9010, sensitivity 0.9090 at specificity 0.8106
- Fusion — AUROC 0.9782, AUPRC 0.9535, sensitivity 0.9852 at specificity 0.8214, ECE 0.0125

The lift survives without the action flags

- Naturalistic test windows (no logged walk, no logged skip): 5936
- Fusion AUROC 0.9781 vs wearable 0.9499, persistence 0.8366, static 0.7588
- Permutation: recent glucose drops AUROC by 0.246; meal and insulin log by 0.183
- HbA1c, BMI, and duration together drop AUROC by only 0.012
- Fusion ECE 0.0125. Static ECE 0.0375.

Lead time, and how often it would buzz

- Usable episodes on test patients: 314. Detected before the crossing: 98.7%.
- Median lead 100 min. Share with at least 30 min lead: 92.6%.
- Alerts while still under 180: 11.89 per patient-day. PPV 0.5759.
- New alert episodes, collapsing consecutive alarms: 2.90 per patient-day.

Trajectory band, and the low-glucose head

- MAE: 4.1 mg/dL at 30 min, 7.2 at 60, 9.2 at 90, 9.8 at 120
- ±1 residual-SD coverage on test: 79.4% to 82.4%
- Hypoglycemia prevalence 0.0127. AUROC 0.9636. AUPRC 0.6606.
- Hypoglycemia sensitivity 0.9487 at specificity 0.9278. ECE 0.0039.

Dashboard: what to click

- Open with python run_demo.py. Twelve synthetic people, ranked by fusion probability.
- Select a row. Compare persistence, static, wearable, and fusion at the same minute.
- Teal line: history the model may use. Dotted line: simulated future it did not see.
- Orange band: fusion trajectory ± 1 residual SD. 180 and 70 are marked.
- Drivers: change in probability when a group is replaced by the training median.

One counterfactual, computed in the open

- Skip the next meal: the log flag is cleared and planned carbohydrates go to zero.
- On 1487 rising test windows, that lowers mean fusion probability by 0.106.
- A 15-minute walk resimulates steps. The flag alone, without new steps, moves mean risk by only 0.017.
- The blue dashed curve is the same physiology with the action applied. History before the decision matches.

Limitations, in the order they matter

- Synthetic physiology. Real CGM, real logging, and real missed doses will be messier.
- Meal-log lead time will shrink the moment people do not log.
- Hypoglycemia AUPRC is 0.66 on a 1.3% event. AUROC alone is the wrong headline.
- Static labs add little once the trace and the log are present.
- No site, no device, no clinician in the loop, no fairness audit across states.

What would change practice, if the signal survived real data

- Put the 2-hour probability next to the HbA1c in a district clinic review
- Use the meal-skip counterfactual as a safety check when a sulfonylurea or insulin was taken
- Use the walk counterfactual as a prescription the person can finish before the peak
- Retune the threshold locally. Specificity 0.80 was a reporting choice, not a guideline.

Reproduce the run

- python run_demo.py
- python -m pytest
- python -m twin.train # writes artifacts/model.joblib and metrics.json
- python -m twin.build_docs
- MIT license. Architecture PDF and Mermaid source are in docs/.

Before this is a submission

- FILL IN team name, college, leader name, phone, and email
- Rename the folder to TeamName_CollegeName
- Make the repository public
- Record this script, unlisted, at least 20 minutes, and paste the URL into the README
- No video URL is claimed in the repository today